

# HDOC – Experiment 29

## Question

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge 2 arrays.
4. Exit

## ❖ Algorithm

- Start.
- Display the menu:
  - Count even, odd, positive, negative, and zero elements.
  - Search for an element using linear search.
  - Merge two arrays.
  - Exit.
- Read the user's choice.
- Use a switch statement:
  - **Choice 1:**
    1. Read the size n.
    2. Read n array elements.
    3. Initialize even, odd, positive, negative, and zero to 0.
    4. Traverse the array.
    5. If an element is zero, increment zero.
    6. Otherwise, check whether it is positive or negative.

7. Check whether it is even or odd.

8. Display all five counts.

- **Choice 2:**

1. Read the size and elements of the array.

2. Read the element to be searched.

3. Traverse the array from beginning to end.

4. If the element is found, display its position.

5. Otherwise, display "Element not found".

- **Choice 3:**

1. Read the sizes of two arrays.

2. Read their elements.

3. Copy the first array into the merged array.

4. Copy the second array after the first array.

5. Display the merged array.

- **Choice 4:** Exit.

- **Default:** Display an invalid-choice message.

- Repeat the menu until the user chooses 4.

- Stop.

### ❖ Test case table

| Test | Choice | Input                      | Expected Output                                         |
|------|--------|----------------------------|---------------------------------------------------------|
| 1    | 1      | n=6, array = 2 -3 0 4 -5 7 | Even = 3, Odd = 3, Positive = 3, Negative = 2, Zero = 1 |

|   |   |                                             |                                                         |
|---|---|---------------------------------------------|---------------------------------------------------------|
| 2 | 1 | n=5, array = 0 0 0 0 0                      | Even = 5, Odd = 0, Positive = 0, Negative = 0, Zero = 5 |
| 3 | 2 | n=5, array = 10 20 30 40 50,<br>search = 30 | Element found at position 3                             |
| 4 | 2 | n=5, array = 10 20 30 40 50,<br>search = 60 | Element not found                                       |
| 5 | 3 | A={1, 2, 3}, B={4, 5}                       | Merged array = 1 2 3 4 5                                |
| 6 | 4 | —                                           | Program terminates                                      |
| 7 | 7 | —                                           | Invalid choice                                          |

### ❖ Program code C

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    int n, n1, n2;
```

```
    int i, key, found;
```

```
    int even, odd, positive, negative, zero;
```

```
    do
```

```
    {
```

```
        printf("\n----- MENU ----- \n");
```

```
printf("1. Count even, odd, positive, negative and zero elements\n");
```

```
printf("2. Search for an element using linear search\n");
```

```
printf("3. Merge two arrays\n");
```

```
printf("4. Exit\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch(choice)
```

```
{
```

```
    case 1:
```

```
    {
```

```
        printf("Enter the size of the array: ");
```

```
        scanf("%d", &n);
```

```
        if(n <= 0)
```

```
        {
```

```
            printf("Invalid array size.\n");
```

```
            break;
```

```
        }
```

```
        int a[n];
```

```
        printf("Enter %d elements: ", n);
```

```
        for(i = 0; i < n; i++)
```

```
        {
```

```
            scanf("%d", &a[i]);
```

```
        }
```

```
even = 0;

odd = 0;

positive = 0;

negative = 0;

zero = 0;


for(i = 0; i < n; i++)
{
    /* Count even and odd */
    if(a[i] % 2 == 0)
        even++;
    else
        odd++;


    /* Count positive, negative and zero */
    if(a[i] > 0)
        positive++;
    else if(a[i] < 0)
        negative++;
    else
        zero++;
}


printf("\nEven elements  = %d\n", even);
printf("Odd elements   = %d\n", odd);
printf("Positive elements = %d\n", positive);
printf("Negative elements = %d\n", negative);
```

```
printf("Zero elements  = %d\n", zero);
```

```
break;
```

```
}
```

case 2:

```
{
```

```
printf("Enter the size of the array: ");
```

```
scanf("%d", &n);
```

```
if(n <= 0)
```

```
{
```

```
printf("Invalid array size.\n");
```

```
break;
```

```
}
```

```
int a[n];
```

```
printf("Enter %d elements: ", n);
```

```
for(i = 0; i < n; i++)
```

```
{
```

```
scanf("%d", &a[i]);
```

```
}
```

```
printf("Enter the element to search: ");
```

```
scanf("%d", &key);
```

```
found = 0;
```

```

for(i = 0; i < n; i++)
{
    if(a[i] == key)
    {
        printf("Element %d found at position %d.\n",
            key, i + 1);
        found = 1;
        break;
    }
}

if(found == 0)
{
    printf("Element %d not found in the array.\n", key);
}

break;
}

case 3:
{
    printf("Enter the size of first array: ");
    scanf("%d", &n1);

    printf("Enter the size of second array: ");
    scanf("%d", &n2);

```

```
if(n1 <= 0 || n2 <= 0)
{
    printf("Invalid array size.\n");
    break;
}

int a[n1], b[n2], merged[n1 + n2];

printf("Enter %d elements of first array: ", n1);
for(i = 0; i < n1; i++)
{
    scanf("%d", &a[i]);
}

printf("Enter %d elements of second array: ", n2);
for(i = 0; i < n2; i++)
{
    scanf("%d", &b[i]);
}

/* Copy first array */
for(i = 0; i < n1; i++)
{
    merged[i] = a[i];
}

/* Copy second array */
for(i = 0; i < n2; i++)
```



```
{
    merged[n1 + i] = b[i];
}

printf("Merged array: ");

for(i = 0; i < n1 + n2; i++)
{
    printf("%d ", merged[i]);
}

printf("\n");

break;
}

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice. Please enter a choice from 1 to 4.\n");
}

} while(choice != 4);

return 0;
}
```

## ❖ Compile, Execute and Evaluate

```
(kali㉿kali)-[~/Desktop/exp29]
$ vi array2.c

(kali㉿kali)-[~/Desktop/exp29]
$ gcc array2.c -o array2

(kali㉿kali)-[~/Desktop/exp29]
$ ./array2

—— MENU ——
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 1
Enter the size of the array: 6
Enter 6 elements: 2 -3 0 4 -5 7

Even elements    = 3
Odd elements     = 3
Positive elements = 3
Negative elements = 2
Zero elements    = 1

—— MENU ——
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 1
Enter the size of the array: 5
Enter 5 elements: 0 0 0 0 0

Even elements    = 5
Odd elements     = 0
Positive elements = 0
Negative elements = 0
Zero elements    = 5
```

```
—— MENU ——
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter the size of the array: 5
Enter 5 elements: 10 20 30 40 50
Enter the element to search: 30
Element 30 found at position 3.

—— MENU ——
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter the size of the array: 5
Enter 5 elements: 10 20 30 40 50
Enter the element to search: 60
Element 60 not found in the array.

—— MENU ——
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 3
Enter the size of first array: 2
Enter the size of second array: 1
Enter 2 elements of first array: 1 2 3
Enter 1 elements of second array: Merged array: 1 2 3

—— MENU ——
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 3
```

```
Enter the size of first array: 3
Enter the size of second array: 2
Enter 3 elements of first array: 1 2 3
Enter 2 elements of second array: 4 5
Merged array: 1 2 3 4 5

—— MENU ——
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge two arrays
4. Exit
Enter your choice: 4
Exiting program...
```

```
(kali@kali)-[~/Desktop/exp29]
$
```